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Commentary

Predicting Choroidal Melanoma Regression after Brachytherapy

Bertil Damato, MD, PhD - *San Francisco, California*

After brachytherapy, some choroidal melanomas regress faster than others. Several studies suggest that these tumors are more likely to metastasize than those that regress slowly. Not all authors support this view, some reporting no correlation between tumor regression and monosomy 3, which is highly lethal.[1] More research is needed to determine whether tumor regression rate after brachytherapy predicts metastasis.

In this issue of *Ophthalmology*, Rashid et al[2] (p. 747) identify clinical features correlating with tumor regression. They confirm that thicker tumors regress more quickly and have created mathematical models demonstrating that tumors regress more slowly as they become smaller. They show that tumors regress more rapidly after ruthenium plaque radiotherapy than with iodine plaques. They describe their statistical methods in detail to allow future authors to adjust for clinical differences between tumors when comparing regression rates between groups (e.g., monosomy 3 vs. disomy 3).

Why should tumor regression rate predict metastatic mortality? Ionizing radiation damages DNA, preventing mitosis, so that cells undergo senescent death. Aggressive tumors with a short lifespan therefore regress more rapidly. This happens with moderate levels of radiation, as with proton beam radiotherapy and other forms of teletherapy.

With brachytherapy, such as plaque radiotherapy, other factors induce tumor regression. Radiation dose diminishes with distance from the radioactive source. To deliver enough radiation to the tumor apex, one therefore must expose the tumor base to very high radiation doses. Such strong radiation disrupts cell membranes, causing autolysis and immediate cell death. Blood vessels also are obliterated, causing tumor infarction. The tumor volume undergoing such immediate destruction is greater with thick melanomas because these need higher radiation doses. The volume of nonsenescent tumor disintegration also is greater with ruthenium plaques, which emit short-range β-radiation, making it necessary to deliver more radiation to the tumor base than iodine plaques, which emit long-range γ-radiation. These insights raise several questions. Do thicker tumors regress more quickly because of shorter cell doubling times or because they undergo more autolysis and infarction? Do large choroidal melanomas metastasize before or after they grow big?

We need to learn more about the pathologic features of irradiated uveal melanomas to treat radiation-induced ocular morbidity better and to improve prognostication. For example, neovascular glaucoma previously was attributed to collateral radiation-induced damage to the iris, ciliary body, and other healthy ocular tissues, but is now known to be caused, in most cases, by what the author has termed the "toxic tumor syndrome," comprising radiation-induced intratumoral vasculopathy, which induces vascular occlusion, tumor ischemia, angiogenic factors, neovascularization, vascular incompetence, hard exudates, macular edema, and exudative retinal detachment. These all resolve if the toxic tumor is removed surgically.[3]

Other phenomena influence tumor regression. Edematous tumor swelling can cause pseudogrowth during the first few months after radiotherapy. Necrotic tumor can squeeze through a break in Bruch's membrane, mimicking growth. This may explain why Rashid et al found collar-stud tumors to regress more slowly. Melanomacrophages can accumulate in great numbers to slow regression, especially in pigmented tumors, as suggested by data from the study by Rashid et al. Extensive fibrosis in irradiated melanomas can reduce regression.

Other changes are important, apart from tumor regression. After radiotherapy, tumors develop inflammatory infiltrates, which may alter the results of genetic analysis, possibly by diluting the melanoma genotype. Prognostic biopsy of choroidal melanoma immediately after proton beam radiotherapy seems reliable, at least with multiplex ligation-dependent probe amplification[4]; however, we need to know how reliability changes over time and how such change varies with radiotherapy type and genetic technique. This need is growing as the scope of genetic prognostication increases because of greater patient expectations and encouraging results from novel therapies for metastatic disease.

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The article by Rashid et al focuses on successful brachytherapy, but Dogrusöz et al[5] found more frequent and more complex chromosomal aberrations in irradiated melanomas. This raises an important question as to whether radiation failure allows or even induces the development of lethal genetic abnormalities in the surviving tumor, actually causing metastatic death. We need studies comparing the genetic profiles of recurrent tumors with those present before treatment. Unless successful in sterilizing the entire tumor, radiotherapy may not be as safe as enucleation after all. When considering brachytherapy, we should be selecting our patients (and surgeons) very carefully—just in case.

References

1. Chiam PJ, Coupland SE, Kalirai H, et al. Does choroidal melanoma regression correlate with chromosome 3 loss after ruthenium brachytherapy? *Br J Ophthalmol*. 2014;98(7):967–971.

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